ME594: 3D Robotics, HW 4
Homework due on 02-09-2025, Topics: Forward Kinematics.

1. **Forward Kinematics of OpenMANIPULATOR-X:** Figure 1 shows the OpenMANIPULATOR-X by Robotis. A video of the robot in action is here: https://youtu.be/B2pnXtooKOg?si=4n-s3DzvyBLluBlA. In the configuration shown in Fig. 1, all the joint angle are set to zero. This configuration is also known as the zero reference configuration. The world frame $o_0x_0y_0z_0$ (not shown) is along the $o_1x_1y_1z_1$ frame (shown in the figure) in the zero reference configuration. The manipulator has four revolute joints: θ_1 along z_1, θ_2 along y_2, θ_3 along y_3, and θ_4 along y_4. These joint axes are shown in the 3D view in Fig. 2 (a). The body frames are $o_ix_iy_iz_i$ where $i = 1, 2, 3, 4$ are at the joint axis and are along the world frame $o_0x_0y_0z_0$ in the zero reference configuration. These body axes are in the 3D view in Fig. 2 (b). The end-effector is the point E in Fig. 1 and shown with a yellow dot in Fig. 2.

   (a) Write down formulae for H_1^0, H_2^0, H_3^0 and H_4^0 as a function of θ_i where $i = 1, 2, 3, 4$.

   (b) Write down a formula for the end-effector's position in the world frame E^0 as a function of the homogenous transformations computed above. You do not need to simplify your answer.

   HINT: To check your formula, you can check the solution for the end-effector for a few special cases that can be deduced by inspecting Fig. 1.

   (a) For $\theta_i = 0$ where $i = 1, 2, 3, 4$ from inspecting Fig. 1, $E^0 = [0.2740, 0, 0.2050]$.

   (b) For $\theta_1 = 90^0$, $\theta_i = 0$ where $i = 2, 3, 4$ from inspecting Fig. 1, $E^0 = [0, 0.2740, 0.2050]$.

   (c) For $\theta_3 = -90^0$, $\theta_i = 0$ where $i = 1, 2, 4$ from inspecting Fig. 1, $E^0 = [0.024, 0, 0.4550]$.

   - **Gradescope submission:** Submit your answers to both questions.
   - **Email submission:** None

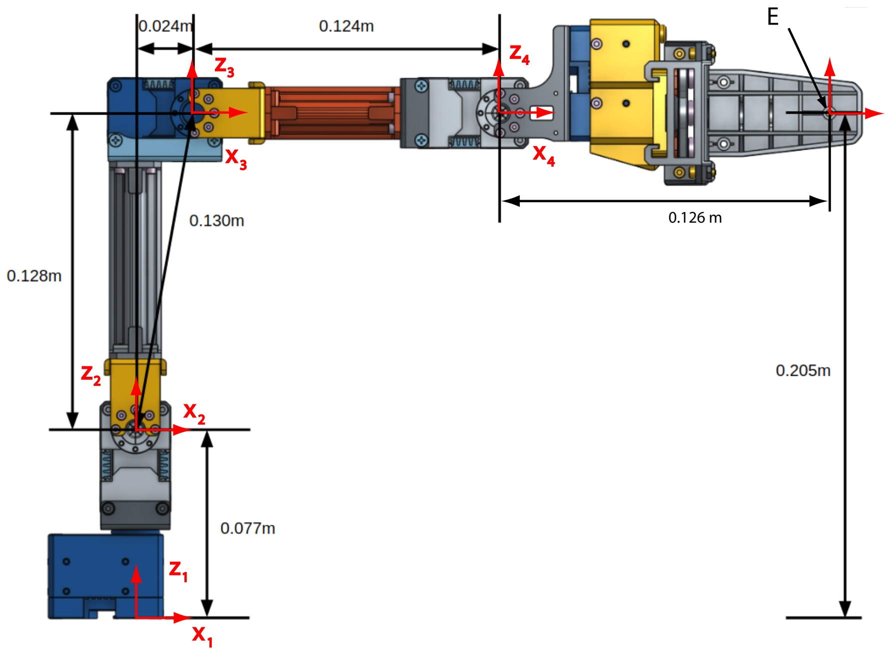

Figure 1: 2D view of the OpenMANIPULATOR-X with joint axis and dimensions.

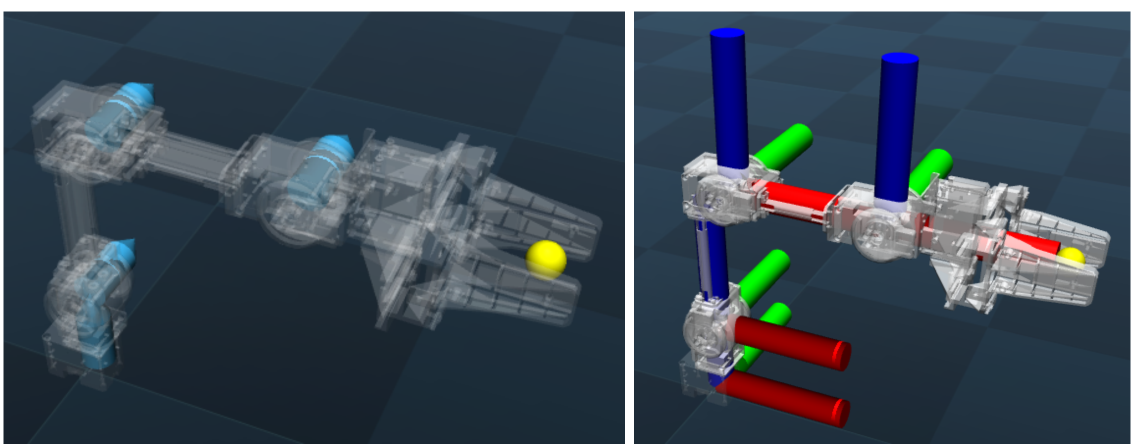

Figure 2: Perspective view of the OpenMANIPULATOR-X showing: (a) the joint axes and (b) the body frames. Note that there are 4 body frames which correspond to the 4 joints. The end-effector is the yellow sphere.

2. **Forward Kinematics of Trossen Robotics WidowX 250 S:** Figure 3 shows the WidowX 250S by Trossen Robotics. A video of the robot in action is here: https://www.youtube.com/watch?v=l_10IHoJNxI. I have provided the wx250s.xml files, the scene.xml and meshes can be downloaded from Github: https://github.com/google-deepmind/mujoco_menagerie/tree/main/trossen_wx250s Create a forward kinematics model of the manipulator. Given the 6 joint angles, q (shown on the right side of the Fig. 3), it should produce the end-effector position and orientation (red sphere). You don't need to submit this part, but it is needed for the next few questions. See **HINT** below for instructions on how to check your answers for test cases.

   (a) What is the end-effector position and orientation (quaternion) for the following joint angles $q = np.array([0.5123, -1.6532, 0.5439, -0.3432, -0.1125, 0.2342]);$?

   (b) Plot the reachable space of the robot (only x, y, z positions of the end-effector). Use the joint range with a reasonable grid size.

   (c) We need to compute the joint angles q that achieve tip position of $[0.4, 0.2, 0.3]$ and tip orientation of $[1, 0, 0, 0]$. Outline a method to achieve this using the solution to the reachable space. Then submit your solution to the best q and report the actual tip position using the forward kinematics.

   HINT: Here are some test cases to check the forward kinematics. For the given q, check the site position/orientation from your code.

   - Test case 1: $q = np.array([0, 0, 0, 0, 0, 0])$, gives site position $[0.4647, 0.0000, 0.3607]$ and site orientation (quaternion) $[1, 0, 0, 0]$.
   - Test case 2: $q = np.array([np.pi/2, 0, 0, 0, 0, 0]);$, gives site position $[0.0000, 0.46470.3607]$ and site orientation (quaternion) $[0.7071, 0, 0, 0.7071]$.
   - Test case 3: $q = np.array([-np.pi/3, 0.5, -0.7, 0.2, 0.3, 0.8]);$ gives site position $[0.2946, -0.4910, 0.3403]$ and site orientation (quaternion) $[0.7618, 0.4400, -0.1900, -0.4359]$.

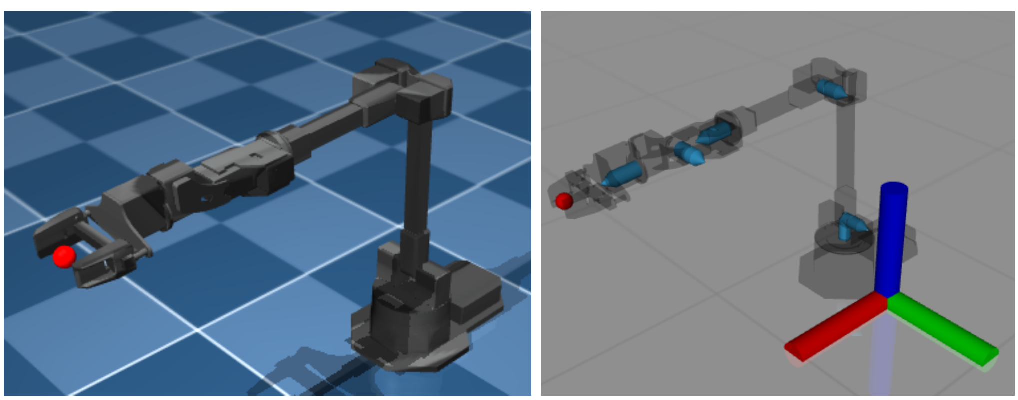

Figure 3: Perspective view of the Trossen WidowX 250 S: The robot has 6 joints. The world frame is at the base of the WidowX but moved to the side in the figure to avoid clutter. The body frames are in the same direction as the world frame and centered at the joints.

- **Gradescope submission:** Submit solutions/plot; no code needs to be submitted.
- **Email submission:** None